

Xinlin Zhuang

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Education

East China Normal University, ME in Computer Science and Technology Sep 2023 – Jun 2026

- GPA: 3.7/4.0
- Honors: **ACL 2025 Best Theme Paper (First-author, 3/8000+ Submissions)**
- Awards: Corporate Scholarship x 2
- Research interests: Efficient training for Large language models (LLMs) and Vision language models (VLMs)

Ocean University of China, BE in Computer Science and Technology Sep 2019 – Jun 2023

- GPA: 3.8/4.0 Rank: 8/269
- Honors: Honorary Bachelor Degree (**Top 0.5%**), Shandong Province Outstanding Graduate (**Top 1%**), University Outstanding Student x 3
- Awards: National Scholarship, First-class University Scholarship x 3, AEON Corporate Scholarship
- Main Courses: Deep Learning, Linear Algebra, Probability Theory

Research Experience

Visiting Student, Department of Computational Biology, MBZUAI – UAE May 2025 – Present

- **Research Focus:** Designing evaluation methods for medical text and image quality, and assessing vision-text alignment to enhance VLM pre-training efficiency. Supervised by Prof. **Eran Segal** and Prof. **Imran Razzak**.

LLM Research Intern, OpenDataLab Team, Shanghai AI Laboratory – China Apr 2024 – Apr 2025

- **Research Focus:** Developing multi-dimensional text quality evaluation methods to enhance LLM pre-training efficiency. Supervised by Prof. **Conghui He**.

Projects

Data Selection for Training Medical Foundation Models Jul 2025 - Present

- Explore innovative data curation methodologies for assessing medical multimodal content quality and optimizing image-text pair alignment to enhance VLM pre-training and fine-tuning efficiency.
- **Technical Stack:** Python, vLLM, LLaMA-Factory.

Data Selection for LLM Medical Reasoning May 2025 - Jul 2025

- Proposed a novel data selection framework combining medical difficulty scores with model influence metrics, achieving equivalent performance with **only 1%** of selected data compared to full-dataset training, and **10%** selection consistently outperforming random selection. *Submitted to AAAI 2026*.
- **Technical Stack:** Python, vLLM, LLaMA-Factory.

Data Selection for Pre-training LLMs Apr 2024 - Apr 2025

- Developed a regression-based data selection framework approximating downstream validation loss, integrating 25 quality metrics including natural signals, importance scores, and model-based ratings for LLM pre-training. Validated across 1B, 3B, and 7B parameter models. **ACL 2025 Best Theme Paper Award**.
- Designed a topic-based data mixing methodology for LLM pre-training using novel large-scale clustering to categorize 600M documents. Demonstrated effectiveness through comprehensive experiments across 1B, 3B, and 7B model scales. *Submitted to AAAI 2026*.
- Developed lightweight BERT-style classifiers via LLM distillation for extracting clean content from diverse HTML sources (forums, e-commerce, articles).
- **Technical Stack:** Python, Apache Spark, LM-Evaluation-Harness.

Automatic Essay Comment Generation for Intelligent Education Dec 2023 - Feb 2024

- Developed the first benchmark for evaluating topic relevance in Chinese K-12 student essays, encompassing relevance classification and automated comment generation. Demonstrated that combining Supervised

Fine-Tuning (SFT) with Direct Preference Optimization (DPO) significantly reduces hallucinations in open-source LLMs for educational assessment tasks. **ACL 2024 Findings.**

- **Technical Stack:** Python, LabelStudio, LLaMA-Factory.

Publications

Meta-rater: A Multi-dimensional Data Selection Method for Pre-training Language Models **ACL 2025 Best Theme Paper**

Xinlin Zhuang, Jiahui Peng, Ren Ma, YinFan Wang, Tianyi Bai, Xingjian Wei, Jiantao Qiu, Conghui He

TOREE: Evaluating Topic Relevance of Student Essays for Chinese Primary and Middle School Education **ACL 2024 Findings**

Xinlin Zhuang, Hongyi Wu, Xinshu Shen, Peimin Yu, Gaowei Yi, Xinhao Chen, Tu Hu, Yang Chen, Yupei Ren, Yadong Zhang, Youqi Song, Binxuan Liu, Man Lan

Harnessing Diversity for Important Data Selection in Pretraining Large Language Models **ICLR 2025 Spotlight (Top 5%)**

Chi Zhang, Huaping Zhong, Kuan Zhang, Chengliang Chai, Rui Wang, Xinlin Zhuang, Tianyi Bai, Jiantao Qiu, Conghui He

Towards Efficient Medical Reasoning with Minimal Fine-Tuning Data Submitted to AAAI 2026

Xinlin Zhuang, Feilong Tang, Haolin Yang, Shixiang Tang, Junjun He, Zongyuan Ge, Ying Qian, Imran Razzak

Topic Over Source: The Key to Effective Data Mixing for Language Models Pre-training Submitted to AAAI 2026

Jiahui Peng*, Xinlin Zhuang*, Ren Ma, Jiantao Qiu, Conghui He

Efficient Pretraining Data Selection for Language Models via Multi-Actor Collaboration **ACL 2025 Main**

Tianyi Bai, Ling Yang, Zhen Hao Wong, Jiahui Peng, Xinlin Zhuang, Chi Zhang, Lijun Wu, Jiantao Qiu, Wentao Zhang, Binhang Yuan, Conghui He

FinDABench: Benchmarking Financial Data Analysis Ability of Large Language Models **COLING 2025**

Shu Liu*, Shangqing Zhao*, Chenghao Jia*, Xinlin Zhuang*, Zhaoguang Long, Jie Zhou, Aimin Zhou, Man Lan

Are U a Joke Master? Pun Generation via Multi-Stage Curriculum Learning towards a Humor LLM **ACL 2024 Findings**

Yang Chen, Chong Yang, Tu Hu, Xinhao Chen, Man Lan, Li Cai, Xinlin Zhuang, Xuan Lin, Xin Lu, Aimin Zhou

CEAMC: Corpus and Empirical Study of Argument Analysis in Education via LLMs **EMNLP 2024 Findings**

Yupei Ren, Hongyi Wu, Zhaoguang Long, Shangqing Zhao, Xinyi Zhou, Zheqin Yin, Xinlin Zhuang, Xiaopeng Bai, Man Lan

Towards Explainable Chinese Native Learner Essay Fluency Assessment: Dataset, Tasks, and Method **EMNLP 2024 Findings**

Xinshu Shen, Hongyi Wu, Yadong Zhang, Man Lan, Xiaopeng Bai, Shaoguang Mao, Yuanbin Wu, Xinlin Zhuang, Li Cai

Skills

Programming Languages: Python, C++

Language: IELTS 8.0 (Reading 9.0, Listening 8.5, Writing 7.0, Speaking 7.0)

Academic Service: Regular reviewer for conferences (*ACL ARR*, *IJCNN*) and associated workshops; Journal reviewer for *Applied Intelligence*.